

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes

Total Marks:20

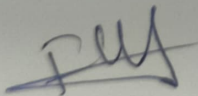
Annexure A

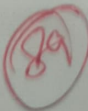
Descriptive Written Test

Code of Conduct:

I 192565026
Shaik Fahed (Name / Reg No) certify that this submission is my original work
and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:







Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

Smart Home Automation System Using Intelligent Agents

A smart home automation system is an Artificial Intelligence (AI) application that automatically controls household devices, such as lights, fans, air conditioners and security systems.

1) Simple Reflex Agent:-

A Simple Reflex Agent makes decisions based only on the current input using predefined if-then rules. It does not store previous information or learn from experience.

Advantages:-

Very fast response

Easy to design and implement

Suitable for simple tasks

Disadvantages:-

Cannot learn user preferences

Cannot handle complex situations

2) Model-Based Agent:-

A model based agent maintains an internal model (memory) of the environment. It remembers previous states and uses them to make better decisions.

3) Goal-Based Agent:-

A Goal-Based Agent selects actions that help achieve specific goals. Instead of simply reacting to sensor input, it plans the best sequence of actions.

4) Utility Based Agent:-

A utility Based Agent chooses the action that provides the maximum overall benefit. It evaluates multiple possible actions and selects the one that best balances comfort, energy efficiency and security.

Robot in an Unknown Maze Using Uninformed Search Strategies :-

Consider a robot placed inside an unknown maze with the objective of finding the exit. Since the robot has no prior knowledge of the maze layout, it must explore the environment using uninformed search algorithms.

1) Uniform Cost Search (UCS) :-

Uniform Cost Search expands the node with the lowest cumulative path cost first. It uses a priority queue to ensure that the cheapest path is always explored before more expensive ones.

Algorithm :-

1. Start from the initial position.
2. Insert the starting node into the priority queue with cost 0.
3. Remove the node with the min cost.

4. If the goal is reached, stop.

5. Otherwise, expand all neighboring nodes and update their costs.

6. Repeat until the exit is found.

Complexity:- Time complexity:- $O(b^{(1+c*/\epsilon)})$

Space complexity:- $O(b^{(1+c*/\epsilon)})$

2) Iterative Deepening Search (IDS):-

Iterative Deepening search combines the advantages of Breadth-First search and Depth-First Search and Depth-First Search.

Algorithm:- 1. Set Depth limit to 0.

2. Perform Depth-First Search up to that limit

3. If the goal is not found, increase the depth limit by one.

4. Repeat until the exit is found.

Complexity:-

Time complexity:- $O(b^d d)$

Space complexity:- $O(bd)$